



MoLeMo Lab

MoJev: Schema-Conditioned Causal Decision Learning with Large Language Models

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For language-model decisions over alternatives defined at inference time, the target is a probability distribution over semantic outcomes. We introduce MoJev, a decision LLM that maps caller-supplied candidates to probabilities through a shared typed head. We formalize Reinforcement Learning for Calibrated Decisions (RLCD) as a typed contextual bandit and derive its policy-gradient solution. Inverting the KL-regularized optimal policy yields Calibrated Decision Preference Optimization (CDPO), which combines grouped Plackett–Luce preference learning with Brier scoring of observed outcomes. TreePacked Attention shares observation computation while separating candidate branches; its per-field predictions feed CDPO in the same training pass. MoJev-Mix aligns preference grades and outcome labels from heterogeneous environments. On 12,000 evaluation decisions, MoJev reaches 93.23% top-one accuracy and 0.79% expected calibration error. Predictive entropy identifies high-error cases for selective prediction. On a measured direct-arylation grid, MoJev-guided search reaches 94.97% mean best yield after 38 experiments, compared with 90.14% for random search. Controlled visual and drone-policy interventions show how the predicted distributions respond to changes in evidence and decision mechanisms. We develop Bayesian and causal formulations that use these distributions for belief updating, information acquisition, and treatment-effect modeling.

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Code: [GitHub](#) [Model](#) [Dataset](#)

1 Introduction

Many language-model decisions concern alternatives defined in the request rather than a fixed label vocabulary. A user may ask which action to take, which hypothesis explains an observation, or which outcome a reaction will produce; in each case, the candidate descriptions determine what the probabilities mean. The prediction is therefore a distribution over a semantic support supplied at inference time. It must assign comparable probabilities across candidates, preserve their meaning under reordering, and match observed outcome frequencies. Such a distribution supports expected-utility decisions, measures uncertainty through entropy, and becomes a belief state for updating when an observation model is available [4; 15; 30].

TypeSafe’s Jev introduced the typed decision interface and *Reinforcement Learning for Calibrated Decisions* (RLCD) as its training direction [6]. We give RLCD a typed contextual-bandit formulation. The policy-gradient solution optimizes decisions on the caller’s candidate set; the KL-regularized optimum then yields a DPO-style direct objective for preference data [24; 34]. Outcome labels add a proper scoring signal for the entire predicted distribution. This derivation connects relative judgments and probability estimation in a single learning problem.

We introduce **MoJev** to implement that formulation. A shared **typed decision head** assigns a contextual utility to each supplied candidate and normalizes the utilities into a distribution; the runtime schema specifies how that distribution is returned. **Calibrated Decision Preference Optimization (CDPO)** fits graded preferences with a grouped ranking likelihood and scores outcome probabilities with the Brier rule. **TreePacked Attention** shares the observation across question and candidate branches while preserving their causal separation, so multiple decision fields are evaluated in one backbone pass. We construct **MoJev-Mix** by aligning preference grades and outcome labels across heterogeneous decision environments and controlling their mixture by source and preference depth.

Joint training of the backbone and decision head reaches 93.23% top-one accuracy and 0.79% expected calibration error on 12,000 evaluation decisions. The same distributions expose high-error cases through entropy, guide sequential reaction-condition search, and respond to controlled changes in visual evidence and drone-control variables. We develop the causal and Bayesian formulations that use these predictions for treatment-effect modeling, belief updating, and information acquisition. Together, the model, learning objective, and packed computation make semantic probability distributions a practical interface between language-model judgment and statistical decision-making.

2 Method

The runtime candidate set determines the model’s output space. We first parameterize a distribution on this space with a shared typed decision head, then derive the RLCD policy-gradient objective and its direct CDPO form. Parallel causal prediction specifies the information flow that evaluates these distributions jointly, leading to the serving computation in Section 2.5.

2.1 Problem formulation

Let H_ϕ denote the language-model backbone and g_ψ its decision head. Class coordinates are local to each request, whose candidate descriptions define the admissible outcomes. Together, the backbone and head map this runtime support to its simplex:

$$F_{\phi,\psi} : \bigcup_{K \geq 2} \mathcal{X} \times \mathcal{Q} \times \mathcal{A}^K \longrightarrow \bigsqcup_{K \geq 2} \Delta^{K-1}.$$

The domain is a union over candidate cardinalities; the codomain is a disjoint union of probability simplices. Token representations supply candidate meaning through parameters shared across schemas. For field f , the model receives

$$\mathcal{S}_f = (x, q_f, A_f), \quad A_f = \{a_{f,1}, \dots, a_{f,K_f}\},$$

and must return

$$\mathbf{p}_f = (p_{f,1}, \dots, p_{f,K_f}) \in \Delta(A_f), \quad p_{f,i} \geq 0, \quad \sum_i p_{f,i} = 1.$$

The candidate text defines semantics; indices only name coordinates of the returned vector. The model’s decision must therefore commute with every permutation $\pi \in S_{K_f}$:

$$F_{\phi,\psi}(x, q_f, \pi A_f) = \pi F_{\phi,\psi}(x, q_f, A_f).$$

MoJev reads one scalar utility $u_{f,i} \in \mathbb{R}$ from each contextual candidate representation and normalizes them:

$$p_{f,i} = \frac{\exp u_{f,i}}{\sum_{j=1}^{K_f} \exp u_{f,j}}.$$

Utilities are identified up to a field-wise additive constant:

$$\text{softmax}(\mathbf{u}_f + c\mathbf{1}) = \text{softmax}(\mathbf{u}_f).$$

The identified quantity is the utility difference

$$\log \frac{p_{f,i}}{p_{f,j}} = u_{f,i} - u_{f,j}.$$

For $K_f = 2$, this is Bradley–Terry [9]. For arbitrary cardinality, it is the Luce choice model. Its random-utility interpretation follows the conditional-logit construction [28]. Repeated selection from the remaining set yields the Plackett–Luce ranking likelihood [2; 3].

2.2 Typed decision head

The typed decision head is the composition of a learned utility kernel with a schema-defined functional. The kernel is shared across runtime schemas; the functional fixes the codomain seen by software.

Low-rank contextual utility kernel. Let z be a tokenized request containing the observation, questions, and candidate descriptions. Section 2.4 specifies its packing and attention mask. The backbone produces one contextual state for every token:

$$H_\phi(z) = (h_1, \dots, h_{|z|}).$$

Mean pooling turns the state, question, and candidate spans into $\bar{h}_x$, $\bar{h}_{q_f}$, and $\bar{h}_{a_{f,i}}$. From the model’s perspective, deciding is a compatibility computation between a contextual query and each candidate representation. The head defines

$$c_f = \bar{h}_x + \bar{h}_{q_f}, \quad M = \frac{W_q^\top W_k}{\sqrt{r}}, \quad r = 512.$$

The utility is

$$u_{f,i} = c_f^\top M \bar{h}_{a_{f,i}}, \quad \text{rank}(M) \leq r.$$

The backbone supplies nonlinear semantic representations; M selects the interaction subspace relevant to decisions. Every schema, question, and cardinality shares this head, so gradients from heterogeneous tasks shape one common decision geometry inside the LLM. Pairwise log-odds become

$$\log \frac{p_{f,i}}{p_{f,j}} = c_f^\top M (\bar{h}_{a_{f,i}} - \bar{h}_{a_{f,j}}).$$

Each log-odds comparison is a context-dependent projection of the difference between two LLM candidate representations.

Permutation equivariance. MoJev sorts candidate text before packing and applies the inverse permutation after scoring. Let S be the canonical sort. For any permutation π ,

$$S(\pi A) = S(A).$$

The packed representation and utility vector coincide. Restoring caller coordinates gives

$$p_\theta(\pi A) = \pi p_\theta(A).$$

This is the permutation-equivariance condition for a neural function whose inputs and outputs are indexed by a set [10]. Attention-based set models provide a complementary construction for learning interactions over unordered inputs [19].

Schema-defined codomain. The decision head is typed by a runtime schema $\sigma = (A, \tau_\sigma)$. Its candidate-dependent part produces a probability law on A ; the type map τ_σ determines the mathematical object returned to the caller:

$$g_\psi(H_\phi(z); \sigma) = \tau_\sigma(\text{softmax}_{a \in A} u_{\phi, \psi}(x, q, a)).$$

Typing constrains the codomain of one shared probability model. Every schema uses the same learned parameters and supplies its own semantic support and return functional.

All serving types are functionals of the same simplex:

Table 1 Schema-defined decision types.

type	support	functional
Noul	$\{0, 1\}$	$p(Y = 1)$
Choice	named finite set	$\arg \max_i p_i$ and $\mathbf{p}$
Score	ordered levels v_i	$\mathbb{E}[v_Y] = \sum_i v_i p_i$

A downstream cost matrix $C(d, y)$ induces the Bayes action

$$d^* = \arg \min_d \sum_{y \in A} C(d, y) p_\theta(y \mid x, q, A).$$

Predictive uncertainty and information acquisition. The decision distribution quantifies uncertainty over semantic outcomes through Shannon entropy [30]:

$$H_\theta(Y \mid x, q, A) = - \sum_{a \in A} p_\theta(a \mid x, q, A) \log p_\theta(a \mid x, q, A).$$

Concentrated probability yields low entropy; competing plausible outcomes yield higher entropy. With natural logarithms, entropy lies between zero and $\log |A|$ nats. Normalized entropy $H_\theta / \log |A|$ expresses uncertainty relative to the maximum for that cardinality. Comparisons across requests also depend on the meaning and granularity of their outcome categories.

This distribution supports a further decision: which evidence should be collected next? Fix the target support and write $c = (x, q, A)$. An information-gathering experiment e produces observation O . Supply an observation likelihood $\ell_e(o \mid y, c)$. Together with MoJev’s prediction, it defines a joint distribution and Bayesian update:

$$r_e(y, o \mid c) = p_\theta(y \mid c) \ell_e(o \mid y, c), \quad r_e(y \mid o, c) = \frac{p_\theta(y \mid c) \ell_e(o \mid y, c)}{\sum_{y'} p_\theta(y' \mid c) \ell_e(o \mid y', c)}.$$

The expected reduction in target uncertainty is

$$\text{IG}(e \mid c) = H_\theta(Y \mid c) - \mathbb{E}_{O \sim r_e(\cdot \mid c)} H_{r_e}(Y \mid O, c).$$

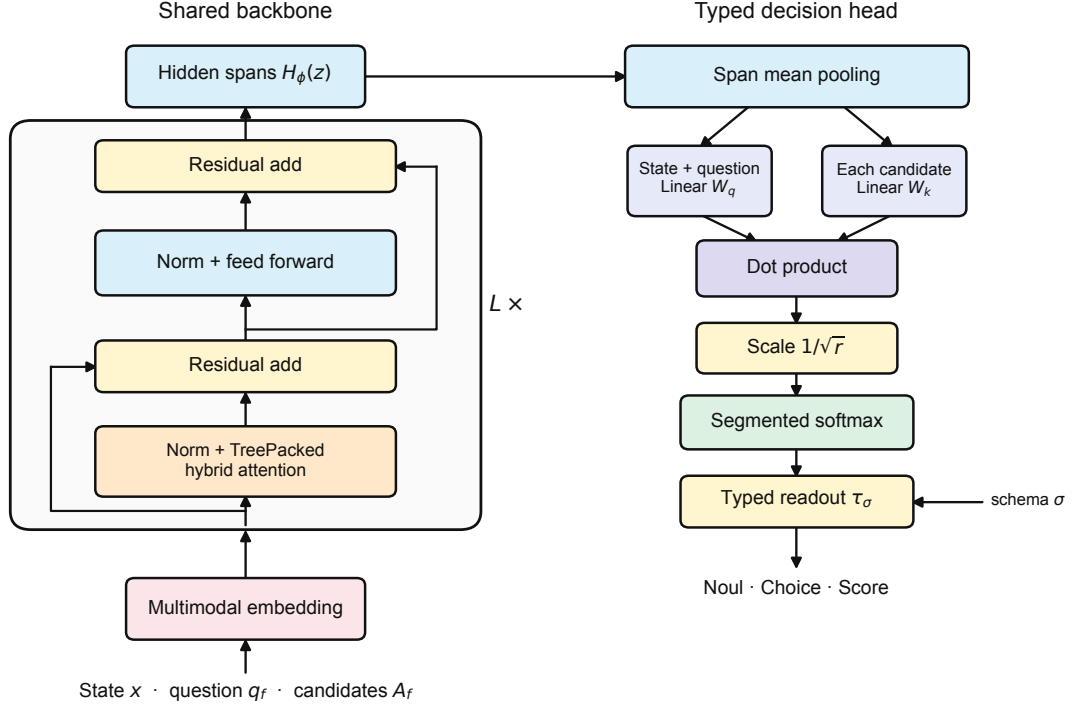


Figure 1 MoJev architecture with a typed decision head. State, question, and candidate descriptions ascend through the shared multimodal backbone under TreePacked visibility. The pooled representations then descend through the typed decision head: query and key projections form scaled candidate utilities, segmented softmax produces a probability law, and a schema-defined readout returns a binary probability, named choice, or expected score. The module layout follows the graphical convention of Transformer architecture diagrams [11].

Maximizing this quantity selects evidence expected to resolve uncertainty about the target, following information-based experimental design [31]. The observation likelihood supplies the update model; the LLM supplies the initial semantic outcome distribution. A conditional predictor used for these updates must agree with the same joint law. For example, a diagnostic decision can compare candidate measurements by their expected reduction in uncertainty over a fixed set of causes. Appendix G derives the entropy bounds and the mutual-information identity.

2.3 From RLCD to CDPO

TypeSafe defines RLCD by its target: typed decisions accompanied by calibrated probabilities [6]. We give that target a one-step reinforcement-learning formulation on the runtime candidate set. Write $c = (x, q, A)$, let $\pi_\theta(a \mid c) = p_\theta(a \mid c)$ be the typed decision policy, and let $Y \in A$ be the observed outcome. Graded feedback g partitions A into $G_1 \succ \dots \succ G_M$; at preference stage m , the remaining actions are $R_m = \bigcup_{t=m}^M G_t$. Restricting and renormalizing π_θ and a full-support reference policy π_0 to R_m gives $\pi_\theta^{(m)}$ and $\pi_0^{(m)}$. For stage reward $r_m(c, a)$, our RLCD objective is

$$J_{\text{RLCD}}(\theta) = \mathbb{E}_{c,g,Y} \left[\frac{1}{M-1} \sum_{m=1}^{M-1} \left\{ \mathbb{E}_{a \sim \pi_\theta^{(m)}} r_m(c, a) - \beta D_{\text{KL}} \left(\pi_\theta^{(m)} \parallel \pi_0^{(m)} \right) \right\} - \lambda_B \|p_\theta(\cdot \mid c) - e_Y\|_2^2 \right].$$

The first term rewards preferred typed actions; KL controls policy movement; the final term scores probabilities against observed outcomes. The one-step policy-gradient identity gives an online solution [34]:

$$\begin{aligned} \nabla_{\theta} J_{\text{RLCD}} = \mathbb{E} \left[\frac{1}{M-1} \sum_{m=1}^{M-1} \mathbb{E}_{a \sim \pi_{\theta}^{(m)}} [(r_m(c, a) - b_m(c)) \nabla_{\theta} \log \pi_{\theta}^{(m)}(a | c)] \right. \\ \left. - \frac{\beta}{M-1} \sum_{m=1}^{M-1} \nabla_{\theta} D_{\text{KL}}(\pi_{\theta}^{(m)} \| \pi_0^{(m)}) - \lambda_B \nabla_{\theta} \|p_{\theta} - e_Y\|_2^2 \right], \end{aligned}$$

where $b_m(c)$ is an action-independent baseline. For samples drawn from a previous policy $\pi_{\text{old}}^{(m)}$, PPO uses the ratio $w_m = \pi_{\theta}^{(m)}(a | c) / \pi_{\text{old}}^{(m)}(a | c)$ and a clipped advantage term [35]:

$$\mathcal{J}_{\text{PPO}}^{(m)} = \mathbb{E}_{a \sim \pi_{\text{old}}^{(m)}} \min \left\{ w_m \hat{A}_m, \text{clip}(w_m, 1 - \epsilon, 1 + \epsilon) \hat{A}_m \right\}.$$

The KL and Brier terms remain differentiable terms in this online objective. CDPO takes the direct route from the same reward-policy relation, following the inversion used in Direct Preference Optimization [24]. For the reward-plus-KL component at each stage, maximizing over the simplex yields

$$\pi_r^{(m)}(i | c) = \frac{\pi_0^{(m)}(i | c) \exp(r_m(c, i) / \beta)}{Z_m(c)}, \quad r_m(c, i) = \beta \log \frac{\pi_r^{(m)}(i | c)}{\pi_0^{(m)}(i | c)} + \beta \log Z_m(c).$$

Substituting the policy ratio for reward into a Luce stagewise-choice model [2; 28] cancels the context-only partition term:

$$P_{\theta}(i | R_m, c) = \frac{\left[\pi_{\theta}^{(m)}(i | c) / \pi_0^{(m)}(i | c) \right]^{\beta}}{\sum_{j \in R_m} \left[\pi_{\theta}^{(m)}(j | c) / \pi_0^{(m)}(j | c) \right]^{\beta}}.$$

For two candidates, this is the DPO-style Bradley-Terry form

$$P_{\theta}(i \succ j | c) = \sigma \left(\beta \log \frac{\pi_{\theta}(i | c)}{\pi_0(i | c)} - \beta \log \frac{\pi_{\theta}(j | c)}{\pi_0(j | c)} \right).$$

With a uniform reference and unit temperature, the multiway stagewise choice probability reduces exactly to the decision head’s restricted softmax, $P_{\theta}(i | R_m, c) = e^{u_i} / \sum_{j \in R_m} e^{u_j}$. The observed top grade at each stage supplies a uniform target over its tied members. Its cross-entropy with this choice probability is the grouped Plackett-Luce term below. Adding the proper Brier score produces the implemented CDPO objective. Appendix B gives the corresponding utility gradient and the treatment of tied grades.

Partial-order likelihood. CDPO trains the LLM directly on a decision distribution. Under a fixed state and question, supervision specifies which candidates should outrank which others. Each training decision assigns an ordinal grade g_i to every candidate, inducing the ordered partition

$$G_1 \succ G_2 \succ \dots \succ G_M, \quad R_m = \bigcup_{t=m}^M G_t.$$

Learning-to-rank literature calls this observation a partitioned preference: partitions are ordered, while candidates inside one partition remain tied [14].

The model turns its utilities into the grouped Plackett-Luce risk. It follows the listwise likelihood principle of optimizing a candidate list as one training instance [13]:

$$\mathcal{L}_{\text{PL}} = -\frac{1}{M-1} \sum_{m=1}^{M-1} \frac{1}{|G_m|} \sum_{i \in G_m} \log \frac{\exp u_i}{\sum_{j \in R_m} \exp u_j}.$$

Per-row normalization gives every decision equal total weight across different order depths.

When only the winner is distinguished,

$$\mathcal{L}_{\text{PL}} = -\log \frac{\exp u_y}{\sum_j \exp u_j},$$

which is multiclass cross-entropy. Pairwise comparison and fully ordered grades recover the Bradley–Terry and Plackett–Luce likelihoods as special cases.

A top-one label supplies $K - 1$ winner-versus-alternative relations. A graded order supplies

$$N_{\text{ord}} = \sum_{m < n} |G_m| |G_n|$$

cross-level relations. These inequalities shape the entire candidate geometry, including relations among non-winning candidates.

Proper probability estimation. The utility ordering identifies the preferred candidate. Probability magnitude requires a proper scoring rule. MoJev reports the full vector $p \in \Delta^{K-1}$. For observed one-hot outcome e_Y , the multiclass Brier loss is

$$\mathcal{L}_{\text{Brier}}(p, Y) = \|p - e_Y\|_2^2.$$

Let $q = P(Y \mid x, q, A)$ be the true conditional distribution. Expanding the expectation gives

$$\mathbb{E}_{Y \sim q} \|p - e_Y\|_2^2 = \|p - q\|_2^2 + 1 - \|q\|_2^2.$$

The Brier regret is exactly

$$\mathbb{E} \mathcal{L}_{\text{Brier}}(p, Y) - \mathbb{E} \mathcal{L}_{\text{Brier}}(q, Y) = \|p - q\|_2^2.$$

Thus $p = q$ is the unique population minimizer. This is the defining identification property of a strictly proper scoring rule [15]. The Brier decomposition further separates reliability, resolution, and outcome uncertainty [27]. The MoJev CDPO risk is

$$\mathcal{R}_{\text{CDPO}}(\theta) = \mathbb{E}_{\mathcal{D}} [\mathcal{L}_{\text{PL}} + \lambda_{\text{B}} \mathcal{L}_{\text{Brier}}], \quad \lambda_{\text{B}} = 1.$$

The grouped preference term shapes utility differences across grade levels. The Brier term has the true outcome distribution as its unique population minimizer. Their sum couples preference structure to outcome prediction through shared parameters; Section 4 measures the accuracy and calibration of the resulting model. [Appendix A](#) derives the joint-risk decomposition and [Appendix B](#) characterizes the stagewise objective.

2.4 Parallel causal prediction with TreePacked Attention

The shared utility kernel compares candidates under the same observation and question. We call this attention mechanism **TreePacked Attention**: each candidate reads its own path, and the common observation is represented once. TreePacked Attention packs the tree into one transformer sequence:

$$z = [x; q_1; a_{1,1}; \dots; a_{1,K_1}; \dots; q_F; a_{F,1}; \dots; a_{F,K_F}].$$

Token visibility. Let S denote the observation-token positions, Q_f the positions of question f , and $C_{f,i}$ the positions of its candidate i . These spans are disjoint. For a query token at position t , define the visible key positions by

$$\mathcal{V}(t) = \begin{cases} S, & t \in S, \\ S \cup Q_f, & t \in Q_f, \\ S \cup Q_f \cup C_{f,i}, & t \in C_{f,i}. \end{cases}$$

Every token can read the entire span to which it belongs. Within-span attention is bidirectional. Across spans, visibility follows the directed state-question-candidate hierarchy. In the following notation, an arrow points from a query span to the spans it can read:

$$x \rightarrow x, \quad q_f \rightarrow \{x, q_f\}, \quad a_{f,i} \rightarrow \{x, q_f, a_{f,i}\}.$$

TreePacked Attention mask. The visibility relation becomes an additive mask applied before attention normalization [11]:

$$M_{ts} = \begin{cases} 0, & s \in \mathcal{V}(t), \\ -\infty, & s \notin \mathcal{V}(t), \end{cases} \quad \text{Attn}(Q, K, V) = \text{softmax}_{\text{row}} \left(\frac{QK^\top}{\sqrt{d_k}} + M \right) V.$$

For one question with two candidates, the block structure is explicit. Rows index query spans and columns index key spans, both ordered as $(S, Q_1, C_{1,1}, C_{1,2})$:

$$M = \begin{pmatrix} 0 & -\infty & -\infty & -\infty \\ 0 & 0 & -\infty & -\infty \\ 0 & 0 & 0 & -\infty \\ 0 & 0 & -\infty & 0 \end{pmatrix}.$$

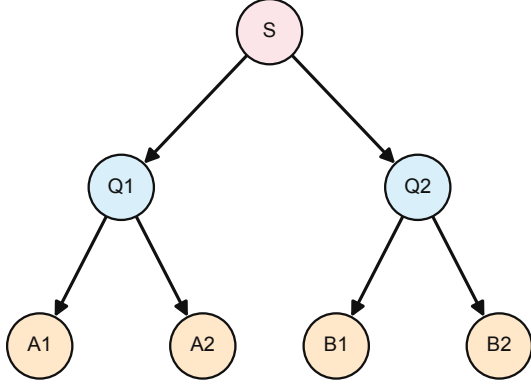
Each entry denotes a rectangular token block. Additional questions contribute separate subtrees: a candidate in field f can read Q_f , while every other question and candidate branch is masked. Observation queries read only observation keys; question queries read the observation and their own question. These restrictions close indirect attention routes through shared ancestors as well as direct routes between siblings.

The mask enforces ancestor visibility for every nonpadding span. Image tokens belong to the observation span and inherit the same visibility rules.

Parallel evaluation. Appendix C states the conditions for packed and separate path evaluations to agree, including positional and recurrent-state requirements.

For attention layers governed by this mask, the ancestor set of every branch is closed under repeated attention: the observation reads itself, a question reads its ancestors, and a candidate reads its ancestors and itself. With token-local remaining operations, induction over layers gives a candidate representation that depends only on its observation-question-candidate path. Packing evaluates these paths together and shares the observation computation [12]. The resulting utilities are normalized separately within each field, yielding the parallel serving computation below.

a



b

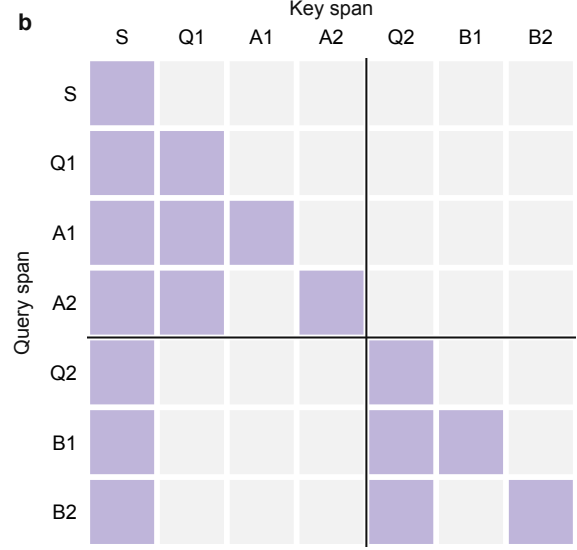


Figure 2 TreePacked attention for one shared state S , two questions Q_1, Q_2 , and two candidates per question. (a) The state-question-candidate tree. (b) Its span-level attention mask: rows are query spans, columns are key spans, and lavender cells are visible. A candidate reads S , its own question, and itself.

Joint training with CDPO. TreePacked Attention and CDPO act on different parts of the same training computation. One masked backbone pass produces the candidate utilities $\mathbf{u}_f$ for every field f . Within each field, CDPO evaluates the grouped preference likelihood on the remaining candidates $R_{f,m}$ at each grade and the Brier score on the full field distribution $\mathbf{p}_f = \text{softmax}(\mathbf{u}_f)$. For the set $\mathcal{F}_{\text{sup}}$ of supervised fields, the request-level loss is

$$\mathcal{L}_{\text{packed}} = \frac{1}{|\mathcal{F}_{\text{sup}}|} \sum_{f \in \mathcal{F}_{\text{sup}}} [\mathcal{L}_{\text{PL}}(\mathbf{u}_f, G_f) + \lambda_B \|\mathbf{p}_f - e_{Y_f}\|_2^2].$$

Every normalization and supervision target stays within its own candidate support. The field losses jointly update the shared decision head and backbone; their gradients meet in the common observation representation, while the tree mask keeps question and candidate branches separate in the forward pass. Thus packed training shares computation without changing the branchwise predictions on which CDPO is defined.

2.5 Efficient serving from parallel causal prediction

Efficient serving follows from the model factorization. A request may contain F questions and $N = \sum_f K_f$ total candidates. MoJev packs the shared state, every question, and every candidate into one tree-masked sequence. The backbone is evaluated once:

$$\{x, (q_f, A_f)_{f=1}^F\} \xrightarrow{\text{pack}} z \xrightarrow{H_\phi(\cdot; M_{\text{tree}})} \{c_f\}_{f=1}^F, \{h_n\}_{n=1}^N.$$

Let $s(n)$ map candidate n to its question. The projected question contexts and candidate states form matrices $C \in \mathbb{R}^{F \times r}$ and $O \in \mathbb{R}^{N \times r}$. All candidate utilities are computed by a gathered row-wise product,

$$u_n = \frac{\langle C_{s(n)}, O_n \rangle}{\sqrt{r}}, \quad n = 1, \dots, N,$$

followed by a segmented softmax over candidates belonging to the same question:

$$p_n = \frac{\exp u_n}{\sum_{m: s(m)=s(n)} \exp u_m}.$$

The segmented distributions complete the parallel causal predictor defined in Section 2.4. Conditioned on the shared state, its joint law over the F field decisions factorizes as

$$P_\theta(Y_1 = i_1, \dots, Y_F = i_F \mid x, \mathcal{S}_{1:F}) = \prod_{f=1}^F p_{f,i_f}.$$

TreePacked Attention realizes this conditional independence inside the transformer: every field reads the same state, and sibling branches remain conditionally isolated. A categorical draw for each field can therefore be taken concurrently from the product law. Probability and argmax requests use the deterministic functionals of that law.

After span pooling and linear projection, utility evaluation costs $O(Nr)$. The complete request cost is

$$T_{\text{MoJev}} = T_{H_\phi}(L) + T_{\text{head}}, \quad N_{\text{decode}} = 0.$$

An autoregressive interface adds a sequential decoding chain; an independent candidate scorer repeats the shared state and question across candidate evaluations. MoJev instead shares the state representation across fields, scores candidates in parallel, and emits no output tokens. Canonical candidate order and restoration of caller order preserve permutation equivariance.

At serving time, MoJev compiles each request into the packed tree z , executes one backbone pass, and applies segmented softmax to materialize the typed outputs.

2.6 Causal Inference

Causal inference. *Potential-outcome distributions.*

Let $T \in \{0, 1\}$ be treatment, X pretreatment state, and $Y \in A$ a typed outcome. Potential-outcome analysis defines a causal effect by comparing the outcomes indexed by treatment for the same unit [8]. MoJev evaluates those treatment-indexed states with shared parameters, question, and semantic support. Define

$$m_k(t, x) = P(Y(t) = a_k \mid X = x), \quad \mathbf{m}(t, x) \in \Delta(A).$$

Holding the candidate schema fixed while changing only the treatment-bearing state gives

$$\hat{\mathbf{m}}_\theta(t, x) = f_\theta(x^{(t)}, q, A)$$

as an LLM parameterization of the conditional potential-outcome distribution.

The model’s conditional distributional effect is

$$\boldsymbol{\tau}(x) = \mathbf{m}(1, x) - \mathbf{m}(0, x).$$

Both terms lie on the same simplex, hence

$$\mathbf{1}^\top \boldsymbol{\tau}(x) = 0, \quad \boldsymbol{\tau}(x) \in T\Delta^{K-1} = \{v \in \mathbb{R}^K : \mathbf{1}^\top v = 0\}.$$

Treatment transports the model’s probability mass among typed outcomes. A value vector $c \in \mathbb{R}^K$ gives the scalar projection

$$\tau_c(x) = c^\top \boldsymbol{\tau}(x).$$

Any scalar estimand defined by outcome values is a projection of this vector.

Identification and g-computation. Under consistency, conditional exchangeability

$$Y(t) \perp T \mid X,$$

and positivity

$$0 < P(T = 1 \mid X = x) < 1,$$

the average effect vector is identified by

$$\mathbf{ATE} = \mathbb{E}_X[\mathbf{m}(1, X) - \mathbf{m}(0, X)].$$

The plug-in estimator is

$$\widehat{\mathbf{ATE}}_g = \frac{1}{n} \sum_{i=1}^n [\hat{\mathbf{m}}_\theta(1, X_i) - \hat{\mathbf{m}}_\theta(0, X_i)].$$

Restricting the expectation to a subgroup yields a CATE vector.

Doubly robust estimation. Let $e(x) = P(T = 1 \mid X = x)$ be the propensity score, the conditional treatment-assignment probability introduced for observational adjustment by Rosenbaum and Rubin [25], and let $\mathbf{Y}_i$ be the one-hot observed outcome. MoJev enters the AIPW estimator

$$\begin{aligned} \widehat{\mathbf{ATE}}_{\text{AIPW}} = \frac{1}{n} \sum_{i=1}^n & \left[\hat{\mathbf{m}}(1, X_i) - \hat{\mathbf{m}}(0, X_i) \right. \\ & + \frac{T_i}{\hat{e}(X_i)} (\mathbf{Y}_i - \hat{\mathbf{m}}(1, X_i)) \\ & \left. - \frac{1 - T_i}{1 - \hat{e}(X_i)} (\mathbf{Y}_i - \hat{\mathbf{m}}(0, X_i)) \right]. \end{aligned}$$

The estimator is consistent when either the propensity or outcome model is correctly specified, the defining double-robustness property [26]. Cross-fitting separates nuisance estimation from effect estimation; this orthogonal-score construction supports valid inference with machine-learned nuisance functions [16]. [Appendix F](#) derives the product remainder and states the convergence conditions.

Causal graphs. A directed acyclic graph supplies the dependency structure for composing MoJev’s conditional predictions. For each discrete node V_j , its candidate schema defines the possible values, and its parent assignment enters the model context. Writing these normalized local factors as $p_{\theta,j}(v_j \mid \text{pa}_j)$ gives

$$p_\theta(v_1, \dots, v_d \mid G) = \prod_{j=1}^d p_{\theta,j}(v_j \mid \text{pa}_j).$$

The factorization defines a joint probability model. Under a causal interpretation with autonomous mechanisms, an intervention fixes the selected nodes and removes their assignment factors [32]. For a set of intervened nodes S ,

$$p_{\theta}(v_{-S} \mid \text{do}(V_S = s), G) = \prod_{j \notin S} p_{\theta,j}(v_j \mid \text{pa}_j) \Big|_{v_S=s}.$$

Marginalizing this distribution gives the intervention’s outcome law. Its entropy measures the remaining outcome uncertainty, which can guide the choice of an additional observation through the information-gain criterion in Section 2.2.

A known graph also provides a reference for probing the learned predictor. Holding a scene fixed, we vary one input node and measure total variation in the output distribution. Comparing these responses with the graph’s mechanism variables tests how the model’s sensitivity aligns with the specified dependencies. The experiment in Section 4.5 uses the drone generator’s collision mechanism as this reference.

Bayesian inference. Bayesian inference treats the candidate set as a set of hypotheses. Let $Z \in A$ denote a hypothesis, with prior $\pi_0(z)$. A schema-conditioned prediction over possible observations supplies a likelihood $L_t(o_t \mid z, o_{1:t-1})$. Sequential conditioning yields

$$\pi_t(z) = \frac{\pi_{t-1}(z) L_t(o_t \mid z, o_{1:t-1})}{\sum_{z' \in A} \pi_{t-1}(z') L_t(o_t \mid z', o_{1:t-1})}.$$

Each update uses the observation probability under the candidate hypothesis. The history argument accommodates dependence between observations. Posterior entropy quantifies the remaining ambiguity over hypotheses, and expected entropy reduction selects the next observation as developed in Section 2.2.

When a predictive factor $q_t(z) = P(Z = z \mid o_t)$ is obtained under a reference prior $\rho(z)$, Bayes’ rule gives a likelihood ratio proportional to $q_t(z)/\rho(z)$. A uniform reference prior makes the predictive factor proportional to the likelihood. This relation specifies how semantic probability predictions enter Bayesian accumulation.

Entropy-weighted probability fusion. For a sequence of semantic predictions over the same candidate set, normalized entropy can regulate the contribution of each distribution. Let $p_t(k)$ denote the prediction at step t . Starting from a prior π_0 , define

$$\pi_t(k) = \frac{\pi_{t-1}(k) p_t(k)^{w_t}}{\sum_{\ell=1}^K \pi_{t-1}(\ell) p_t(\ell)^{w_t}}, \quad w_t = 1 - \frac{H(p_t)}{\log K}.$$

This tempered product-of-experts rule adds entropy-weighted log odds:

$$\log \frac{\pi_T(k)}{\pi_T(\ell)} = \log \frac{\pi_0(k)}{\pi_0(\ell)} + \sum_{t=1}^T w_t \log \frac{p_t(k)}{p_t(\ell)}.$$

For positive probabilities, cancellation of the normalization constants gives this identity. Entropy controls the strength of each prediction’s contribution in log-odds space, with a uniform prediction contributing zero. The fusion rule operates on predictive distributions; the Bayesian construction above specifies observation-likelihood factors.

3 Data Construction

CDPO determines the supervision that each source must provide. We construct rows with a common candidate support for preference grades and observed outcomes, then control the training measure through source-level mixing.

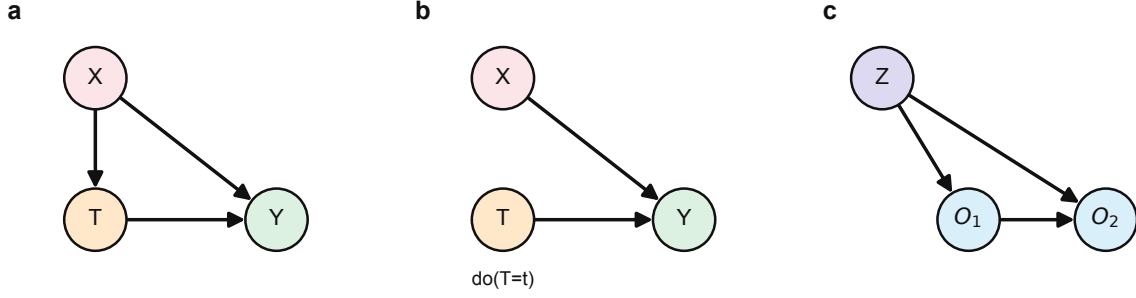


Figure 3 Graphical structures used in distributional inference. (a) Pretreatment state X confounds treatment T and outcome Y ; g-computation and AIPW adjust for X . (b) $\text{do}(T=t)$ removes the assignment edge into T . (c) A hypothesis Z and observation history O_1, O_2 define sequential Bayesian likelihoods. Panel c is a dependency graph for evidence updating.

3.1 Data form induced by CDPO

The CDPO objective determines the row schema. Grouped Plackett–Luce requires an ordered partition of the runtime candidates, while Brier requires one realized outcome on the same support. The atomic observation is

$$D_n = (x_n, q_n, A_n, g_n, y_n), \quad A_n = \{a_{n,1}, \dots, a_{n,K_n}\}, \quad y_n \in A_n,$$

with a grade vector aligned to the candidates,

$$g_n \in \{0, \dots, L_n\}^{K_n}, \quad g_{n,y_n} = \max_i g_{n,i}.$$

The Plackett–Luce term reads (A_n, g_n) , and the Brier term reads (A_n, y_n) . Candidate text defines the local coordinates of the output simplex. The grade vector supplies relative utility, and the observed outcome anchors probability magnitude. Keeping both signals in one row makes the two CDPO terms optimize one conditional distribution.

3.2 Selecting data sources

A usable CDPO source yields a state x with at least two admissible candidates and an observed winner $y \in A$ determined by the source dynamics. Recoverable geometry over the remaining candidates adds Plackett–Luce information.

Preference content is measured by the number of distinct grade levels $|\text{Im } g|$. Binary supervision has two levels. A source with three or more levels also orders some incorrect candidates, giving CDPO constraints inside the losing set.

MoJev-Mix applies this criterion to 18 deterministic Open-Jev generators [7]. Their state transitions supply the winner y . When the environment exposes a distance or rule order $d_x(a)$, it is quantized into grades

$$g_x(a) = Q_x(-d_x(a)), \quad g_x(y) = \max_{a \in A} g_x(a).$$

Graph distance is used directly. Palette distance, ordinal displacement, and rule priority are mapped to four grade bands. A source carrying only the winner receives the flat grading $g(y) = 3$ and $g(a) = 0$ for the remaining candidates. This fallback keeps the outcome valid for Brier and reduces the Plackett–Luce term to multiclass cross-entropy.

3.3 Cleaning and deterministic conversion

Each source row is converted to a common representation. Structured states are serialized into the context, candidate values remain human-readable strings, and the source target is resolved to the candidate with maximal target weight. The converter normalizes the calibration split into validation and retains the original test and OOD boundaries.

Schema validation enforces a nonempty context, at least two candidates, membership of y in A , and exact alignment between g and A . Candidate sets wider than 64 are filtered during conversion. A geometry-specific grader is accepted only when it assigns every candidate; incomplete metadata falls back to flat grading. These rules are implemented in the deterministic converter.

The same schema carries multimodal states. An image marker occupies the state field x , while (A, g, y) retains the identical CDPO representation.

3.4 Mixture construction

The conversion produces 1,271,770 valid rows. Two sources contribute 83% of this pool, so direct concatenation makes source frequency the dominant training signal. Data-mixture research treats source weights as part of the learning problem: DoReMi optimizes domain proportions [17], while DSIR selects examples by importance resampling toward a target distribution [18]. MoJev-Mix uses a deterministic rule tied to CDPO supervision depth. Rows are grouped by source and preference depth, then capped by

$$c(D) = \begin{cases} 60,000, & |\text{Im } g_D| \geq 3, \\ 20,000, & |\text{Im } g_D| = 2. \end{cases}$$

Sources below the cap are preserved in full. Oversized buckets are ranked by a stable hash of the seed, context, and candidate set; the first $c(D)$ rows are retained. A second stable hash fixes output order. The same input and seed therefore generate the same mixture independently of file traversal order.

The final mixture contains 507,974 rows, with 205,084 assigned to training. This construction changes painting geometry from 67.2% to 39.0% of the training measure and raises the share of rows with at least three grades from 28.5% to 38.8%. Validation tracks CDPO model selection and calibration; test retains observed source families; OOD evaluates held-out generators.

4 Results

We evaluate the learned decision distribution through prediction accuracy and calibration, examine its response to candidate transformations, and measure the latency of the packed serving path. We then test sequential reaction-condition search, multimodal intervention, and decision-making in a drone-control environment.

4.1 Training setup

We jointly optimize all 854,036,544 parameters of Qwen3.5-0.8B and the decision kernel as one decision LLM. The empirical-risk minimization uses one epoch, effective batch size 192, learning rate 10^{-5} , a 16,384-token state truncation limit, and $\lambda_B = 1$. The Qwen3.5 backbone supports 262,144 tokens natively and is extensible to 1,010,000 tokens.

4.2 Main result

On 12,000 evaluation decisions sampled from the test and OOD files:

The invariance experiments agree with the model symmetries: reordering candidates or padding the state leaves the decision unchanged, and replacing a distractor preserves the selected candidate. The held-out-

Table 2 Predictive accuracy and calibration.

metric	result
top-one accuracy	93.23%
expected calibration error	0.79%

generator split tests a concrete source shift; distribution-shift benchmarks identify this separation between training domains and deployment domains as a distinct evaluation axis [29].

4.3 Serving efficiency

The packed model path evaluates every field in a request jointly. The released serving measurements are:

Table 3 Serving performance.

serving quantity	result
packed inference, batch 8	34.5 ms
end-to-end HTTP latency, p50	77 ms
autoregressive output tokens	0
maximum served/in-process probability difference	0.0

The zero probability difference shows that HTTP serving transports the same segmented-softmax decision function. Latency is governed by the single packed backbone evaluation in Section 2.5, with no autoregressive decode phase.

4.4 Multimodal intervention

The multimodal LLM accepts image tokens in the state branch. Replacing a gray control image with a cat photograph while preserving the textual question and outcome simplex produces:

Table 4 Effect of visual evidence on predicted cat probability.

support	gray $P(\text{cat})$	cat image $P(\text{cat})$	effect
cat, dog	0.471	0.786	+0.315
cat, dog, car, other	0.264	0.528	+0.264

This controlled intervention shows the same decision interface operating on multimodal state representations: visual evidence moves probability mass while the semantic output coordinates remain fixed. Figure 4 gives a second example with a tabby photograph and a same-size flat-gray control. The question and candidate set are held constant while only the visual state changes.

4.5 Reaction condition optimization

We test MoJev on the direct-arylation benchmark for Bayesian reaction optimization introduced by Shields et al. [33]. Its experimentally measured design space contains 1,728 combinations of 12 phosphine ligands, four bases, four solvents, three concentrations, and three temperatures. The measured yields come from the authors’ [original reaction table](#).

Each search run begins with eight randomly selected experiments. In each of ten rounds, MoJev reads the observed condition–yield pairs, assigns choice probabilities to the remaining conditions, and selects three experiments from the twelve highest-ranked candidates. The returned yields are looked up in the measured table. A random-search control starts from the same eight experiments and samples three remaining conditions per round. We evaluate five paired seeds.

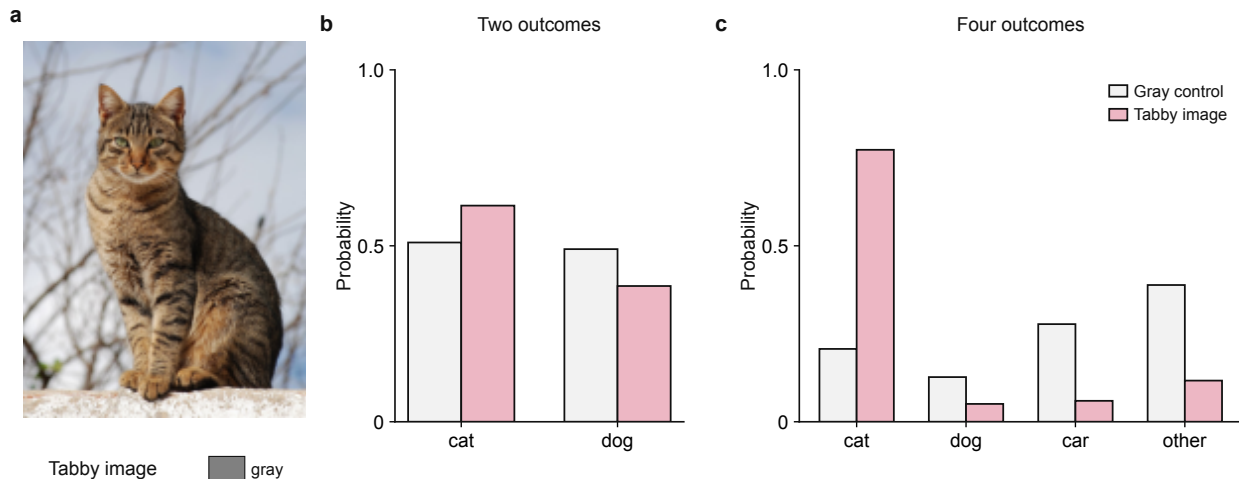


Figure 4 Multimodal state intervention with fixed question and candidate support. (a) The tabby photograph and a same-size flat-gray control. (b,c) Predicted probability distributions for two- and four-outcome choice questions. Replacing gray with the photograph changes $P(\text{cat})$ from 0.509 to 0.614 in b and from 0.207 to 0.773 in c. Photograph: *Cat November 2010-1a.jpg*, Alvesgaspar, Wikimedia Commons, CC BY-SA 3.0; scaled for display.

After 26 experiments, the mean best observed yield is 94.53% for MoJev-guided search and 83.55% for random search. After 38 experiments, the corresponding yields are 94.97% and 90.14% (Figure 5).

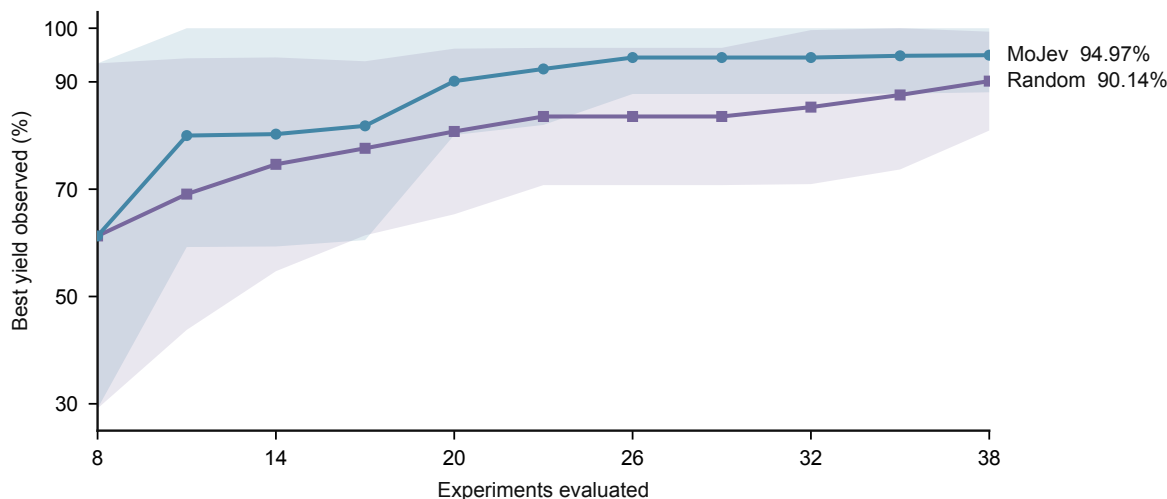


Figure 5 Closed-loop search over the measured direct-arylation grid. Lines show mean best observed yield across five paired seeds; shaded bands show ± 1 sample standard deviation, clipped to the 0–100% yield range. Both methods begin with eight identical experiments and evaluate three new conditions per round; MoJev samples three from its twelve highest-ranked untested conditions.

4.6 Applications in Causal Inference

We study MoJev’s probability predictions in a drone-control environment with a known collision mechanism. The application experiments share one test scene; a separate held-out evaluation measures how well entropy identifies unreliable predictions.

Causal inference. The drone environment defines collision risk through stopping distance and time-to-collision rules. We hold the scene fixed and vary the imminent-collision threshold across eight values from 0.1

to 32 seconds, evaluating the complete risk distribution at each setting. This constructs a distribution-valued response curve for a policy parameter: each coordinate tracks the predicted probability of one risk category.

We compare policy variables by the maximum total-variation distance from the original prediction over two prescribed settings. The table below reports these responses alongside their roles in the generating mechanism. Appendix H specifies the experimental inputs.

Causal graphs. The generator provides a reference graph for interpreting the measured responses. We use failed-scan count as a control because it enters target-loss handling, outside the collision-risk mechanism.

Table 5 Sensitivity to variables in the drone-control mechanism.

Variable	Path in the generating mechanism	Total-variation response
Imminent-collision threshold	Risk classification	0.0216
Braking deceleration	Stopping distance $\rightarrow$ risk	0.0177
Caution threshold	Risk classification	0.0137
Safety margin	Stopping distance $\rightarrow$ risk	0.0062
Failed-scan count	Target-loss control	0.0059
Reaction time	Stopping distance $\rightarrow$ risk	0.0045

The strongest responses follow the braking and risk-classification pathways in the reference graph. Safety margin and reaction time produce changes close to the control. The response profile therefore distinguishes the dominant mechanism variables more clearly than the weaker ones in this scene.

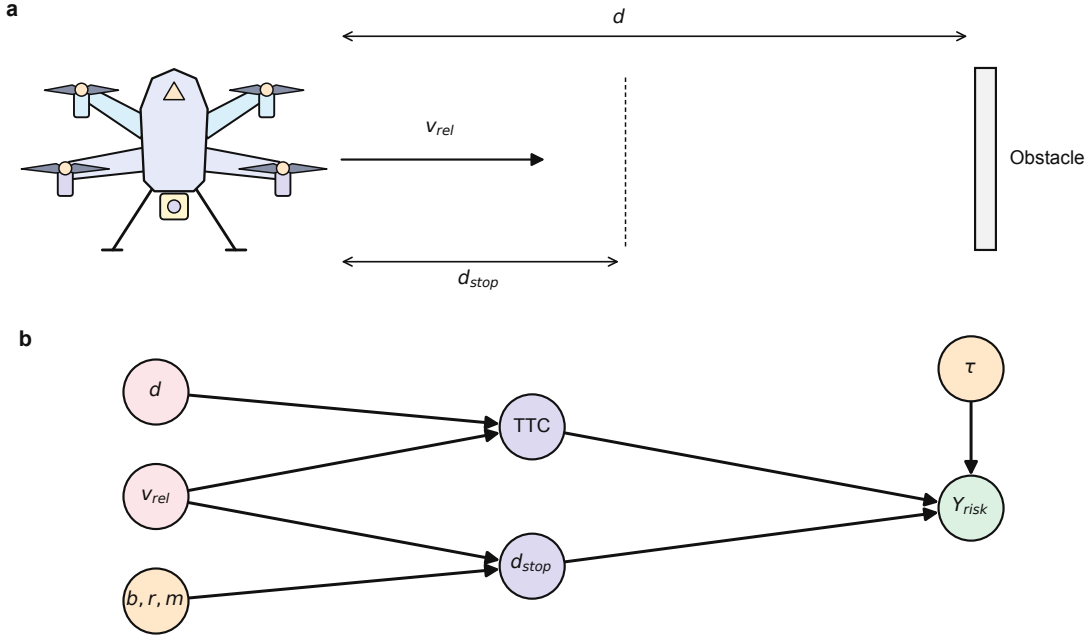


Figure 6 Drone collision-risk mechanism. (a) An action schematic with a front-view quadcopter icon: the drone approaches an obstacle with relative speed v_{rel} , available distance d , and stopping distance d_{stop} . The dashed line marks the illustrated stopping point. (b) The generating dependencies: d and v_{rel} determine time to collision (TTC), while braking b , reaction time r , and margin m determine d_{stop} ; threshold τ enters risk classification. Failed-scan count belongs to the separate target-loss rule.

Bayesian inference. The sequential-inference experiment compares equal-weight and entropy-weighted accumulation from a uniform belief over risk categories. Both conditions receive the same sequence of predictions

from the scene variants specified in Appendix H. This paired design isolates the effect of entropy weighting on the accumulated distribution.

The comparison tracks how each prediction changes the accumulated category probabilities. Equal-weight accumulation gives every factor unit weight; entropy weighting discounts ambiguous factors according to Section 2.6. Both conditions operate on the same outcome space throughout the sequence.

Uncertainty measurement. We assess predictive entropy on a separate 400-example test subset and an OOD split. Entropy is normalized by the logarithm of the number of valid candidates. We rank examples by increasing entropy and measure accuracy at three retained coverage levels.

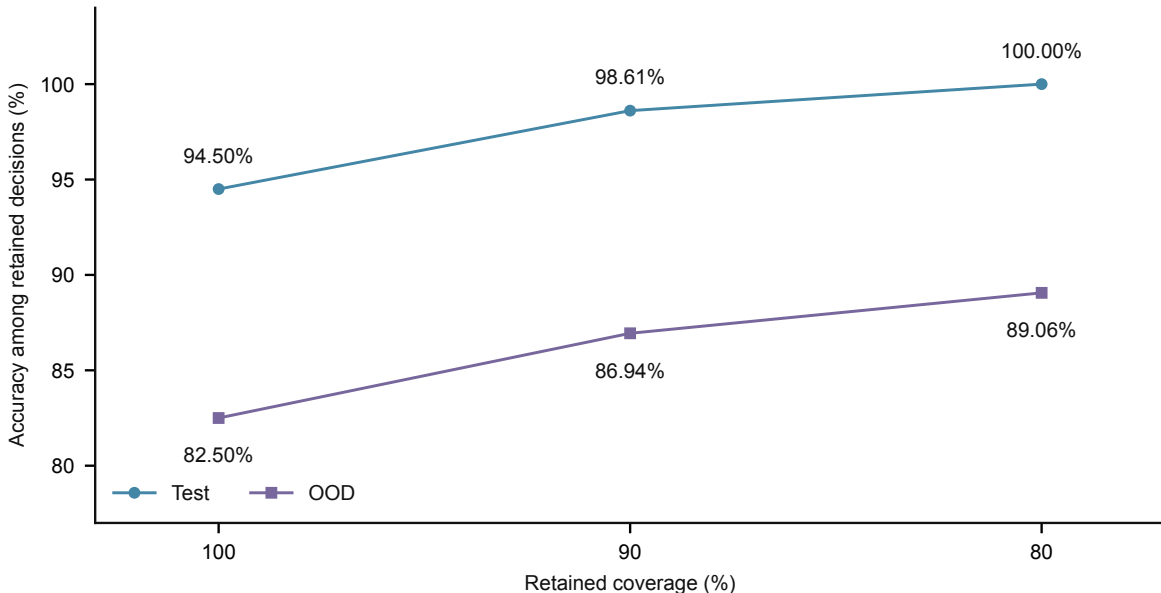


Figure 7 Accuracy at three retained coverage levels after ranking decisions by normalized predictive entropy. Error-detection AUROC is 0.9750 on test and 0.6819 on OOD.

All observed test errors fall in the highest-entropy fifth of predictions. Entropy therefore provides a strong signal for selecting reliable judgments in distribution. Selection also improves OOD accuracy, while the lower AUROC indicates weaker separation between correct and incorrect predictions under shift.

5 Related Work

Preference learning supplies reward signals from comparisons of model behavior [23]. Policy gradients optimize expected return [34], and PPO controls sampled policy updates through a clipped surrogate [35]. The modern LLM pipeline trains reward models on ranked generations and optimizes a language model against the learned reward [22]. Direct Preference Optimization inverts the KL-regularized optimal-policy relation to fit preferences without sampled policy updates [24]. LLaMA-Berry’s Pairwise Preference Reward Model applies Bradley–Terry likelihood to reasoning-path comparison [1; 9]. CDPO applies the reward-to-policy inversion to the typed, variable-size candidate support, extends preference fitting to graded lists, and scores the full outcome distribution.

The Plackett–Luce family defines distributions over choices and rankings [2; 3]. ListMLE turns its likelihood into a listwise learning objective [13], and partitioned-preference models extend it to ordered groups with unresolved within-group order [14]. CDPO uses graded partitions to supervise a shared contextual utility kernel.

Neural calibration work establishes accuracy and confidence quality as distinct statistical properties [4]. Proper-scoring theory characterizes the conditions under which a forecast distribution is truthfully identified

[15]. CDPO incorporates the Brier score [5] into empirical risk minimization, making conditional probability estimation part of representation learning.

Neural functions over sets require permutation symmetry [10], with attention providing a general interaction operator for set-structured inputs [19]. Pointer networks establish variable-size output dictionaries selected from the input [20]. Transformer masks specify directed information access [11], sequence packing preserves independent-example semantics inside a shared accelerator sequence [12], and IO-aware exact attention reduces memory movement during the resulting backbone pass [21]. MoJev uses TreePacked Attention to organize the state-question-candidate graph for parallel causal prediction.

TypeSafe introduced RLCD with Jev as a training direction for typed decisions and calibrated probabilities [6]. Open-Jev supplies public environments for this interface [7]. MoJev trains a fully parameterized LLM on the interface with CDPO’s grouped-preference and proper-scoring objective.

Potential-outcome methods define causal effects through counterfactual outcome distributions [8]. Propensity scores encode the treatment-assignment mechanism [25], doubly robust estimators combine assignment and outcome models [26], and orthogonal scores with cross-fitting connect machine-learned nuisance functions to treatment-effect inference [16]. MoJev supplies a variable-support multiclass outcome model and represents treatment effects in the simplex tangent space.

Language-model data selection makes the training measure explicit through domain reweighting [17] or importance resampling [18]. MoJev-Mix fixes that measure deterministically from source identity and CDPO preference depth.

6 Conclusion

We presented MoJev, an open LLM that predicts distributions over semantic candidates defined at inference time. Its typed decision head gives heterogeneous tasks a shared utility representation, CDPO combines graded preferences with outcome supervision, and TreePacked Attention organizes the computation for parallel prediction. Training on MoJev-Mix demonstrates that this formulation can deliver accurate decisions and calibrated probabilities within a single model.

The experiments show how the predicted distribution supports decisions about the model’s own outputs and actions in external environments. Entropy identifies errors at AUROC 0.975 on 400 test examples, with all 320 retained predictions correct at 80% coverage. MoJev-guided reaction search reaches a mean best measured yield of 94.97% after 38 experiments, compared with 90.14% for random search. Drone-control probes measure how changes in policy variables redistribute risk probabilities, and entropy-weighted fusion uses predictive uncertainty to regulate sequential updates. Together with the Bayesian and causal formulations, these results connect language-model judgments to reasoning over outcomes, evidence, and interventions.

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Appendix: Mathematical Properties

The results below fix the observation, question, and candidate support unless an expectation over observations is explicit. Write $p^\star$ for the conditional outcome distribution. Population risks are defined over the probability simplex; a parameterized model attains their minimizers when its hypothesis class represents those distributions.

A CDPO risk and probability estimation

Proposition A.1 (winner-only supervision). Suppose the observed winner $Y \sim p^\star$ forms the unique top grade and all remaining candidates share the bottom grade. For $\lambda_B > 0$, the conditional CDPO excess risk satisfies

$$R(p) - R(p^\star) = D_{\text{KL}}(p^\star \| p) + \lambda_B \|p - p^\star\|_2^2.$$

Proof. The single preference stage is $-\log p_Y$. Its expected value is $H(p^\star) + D_{\text{KL}}(p^\star \| p)$. The expected Brier score is $\|p - p^\star\|_2^2 + 1 - \|p^\star\|_2^2$. Subtracting the risk at $p^\star$ cancels both constants. Consequently the unique minimizer is $p^\star$, and

$$\|p - p^\star\|_2^2 \leq \frac{R(p) - R(p^\star)}{\lambda_B}.$$

For boundary distributions, the formula uses the usual extended-value convention for KL divergence; finite-logit softmax models approach boundary optima by limits.

Proposition A.2 (graded supervision). Let $A(p) = \mathbb{E}_g[\mathcal{L}_{\text{PL}}(p, g)]$ be the conditional expected graded loss, with finite $A(p^\star)$. Then

$$R(p) - R(p^\star) = A(p) - A(p^\star) + \lambda_B \|p - p^\star\|_2^2.$$

If the grade-generating distribution satisfies the compatibility condition $p^\star \in \arg \min_p A(p)$, the same uniqueness and error bound follow. More generally, define

$$\delta = A(p^\star) - \inf_p A(p).$$

For any ε -optimal joint-risk solution $\hat{p}$,

$$\|\hat{p} - p^\star\|_2^2 \leq \frac{\delta + \varepsilon}{\lambda_B}.$$

Proof. The decomposition follows by cancelling the outcome-dependent Brier constant. Since $R(\hat{p}) \leq R(p^\star) + \varepsilon$ and $A(\hat{p}) \geq \inf_p A(p)$, rearrangement gives the bound. Compatibility sets $\delta = 0$. This quantity expresses the alignment required between graded preferences and outcome probabilities.

B The grouped preference objective

For a fixed context and remaining set R_m , consider the reward-KL component

$$\max_{\pi \in \Delta(R_m)} \sum_{i \in R_m} \pi_i r_i - \beta \sum_{i \in R_m} \pi_i \log \frac{\pi_i}{\pi_{0,i}}.$$

The Lagrangian first-order condition is $r_i - \beta(\log(\pi_i/\pi_{0,i}) + 1) + \eta = 0$. Normalizing gives $\pi_i^* = \pi_{0,i}e^{r_i/\beta}/Z_m$. Thus every reward difference has the policy representation

$$r_i - r_j = \beta \log \frac{\pi_i^*/\pi_{0,i}}{\pi_j^*/\pi_{0,j}}.$$

Under a Luce preference model, substitution removes the unknown normalizer Z_m . For a uniform reference and $\beta = 1$, the resulting choice probability is precisely the restricted softmax of the learned utilities. The binary case gives Bradley–Terry; repeated choices over remaining sets give Plackett–Luce. This is the reward-elimination step linking the policy-gradient objective to direct preference fitting.

For each stage, define the restricted choice distribution and its target by

$$p_i^{(m)} = \frac{e^{u_i}}{\sum_{j \in R_m} e^{u_j}}, \quad t_i^{(m)} = \frac{\mathbf{1}\{i \in G_m\}}{|G_m|}, \quad i \in R_m.$$

The implemented objective is an average of stagewise cross-entropies:

$$\mathcal{L}_{\text{PL}} = \frac{1}{M-1} \sum_{m=1}^{M-1} H(t^{(m)}, p^{(m)}).$$

Differentiating log-sum-exp gives the exact utility gradient

$$\frac{\partial \mathcal{L}_{\text{PL}}}{\partial u_j} = \frac{1}{M-1} \sum_{m: j \in R_m} (p_j^{(m)} - t_j^{(m)}).$$

Each stage has positive-semidefinite Hessian $\text{diag}(p^{(m)}) - p^{(m)}p^{(m)\top}$, embedded in the full candidate coordinates. Thus the loss is convex in the utility vector and invariant to adding a common utility offset. This statement concerns the utility coordinates; the backbone parameterization determines the optimization landscape in model parameters.

With singleton groups, the sum of stage losses is the ordinary PL negative log-likelihood, and the factor $1/(M-1)$ normalizes by ranking depth. With a unique winner and one losing group, it is cross-entropy. With tied groups, the formula defines a uniform-target stagewise surrogate. The exact probability of a partitioned ranking under PL is

$$P_{\text{PL}}(G_1 \succ \cdots \succ G_M) = \sum_{\pi \in \Pi(G)} \prod_{j=1}^{K-1} \frac{e^{u_{\pi_j}}}{\sum_{\ell=j}^K e^{u_{\pi_\ell}}},$$

where $\Pi(G)$ contains all total orders compatible with the groups [14]. This expression marginalizes within-group order. The stagewise surrogate assigns equal target mass to members of the current group, giving the explicit gradient above.

C TreePacked Attention equivalence

Proposition C.1 (pathwise equivalence). Compare a packed tree evaluation with separate evaluations of each observation–question–candidate path. Assume that (i) every operation mixing token positions respects the tree ancestry relation; (ii) corresponding tokens use identical embeddings and position-dependent attention terms in both evaluations; (iii) all remaining operations act independently at each token; and (iv) inference is deterministic. Then every candidate has identical final hidden states and utility in the two evaluations.

Proof. At layer zero, corresponding token representations agree by assumption (ii). Suppose they agree before a layer. For any token, its visible ancestor and own-span tokens are the same in both evaluations.

The corresponding queries, keys, values, and positional terms therefore agree, so attention normalization and the weighted sum agree. Token-local operations preserve equality. Induction gives equality at every layer, and identical pooling and projections give identical utilities.

For a fixed packing and position assignment, the same argument gives the structural derivative property

$$\frac{\partial h_{f,i}}{\partial e(a_{g,j})} = 0, \quad (g,j) \neq (f,i).$$

The assumptions apply to every layer of the backbone. For recurrent layers, each branch must inherit its parent’s recurrent state and continue with a branch-local state; convolution histories require the same branching rule. RoPE positions must preserve the relative positions of visible token pairs. These requirements extend the mask-level construction to a complete backbone equivalence statement. In the current Qwen3.5 integration, the four-dimensional tree mask controls full-attention layers; its recurrent-mask adapter returns no mask for a four-dimensional input. Applying Proposition C.1 to that hybrid backbone requires branch-local recurrent and convolution states.

The product law in Section 2.5 defines the model’s joint prediction. Independent categorical random draws from its field marginals realize that law. Pathwise computational isolation specifies which input tokens determine each utility.

Lemma C.2 (candidate permutation equivariance). Assume candidates have unique canonical keys and deterministic preprocessing. Let S_A map caller coordinates to sorted coordinates, and let G be the deterministic scorer of the canonical sequence. Then $F(A) = S_A^{-1}G(S_AA)$ satisfies

$$F(\pi A) = \pi F(A).$$

Proof. Sorting gives $S_{\pi A}\pi A = S_AA$ and $S_{\pi A}^{-1} = \pi S_A^{-1}$. Substitution proves the result. Duplicate descriptions can be handled by an order-independent tie rule or by assigning equal predictions to indistinguishable candidates.

D From probability error to typed decisions

Proposition D.1 (score error). For outcome values $v \in \mathbb{R}^K$, let $s(p) = v^\top p$. For any constant a ,

$$|s(p) - s(p^*)| = |(v - a\mathbf{1})^\top (p - p^*)| \leq \|v - a\mathbf{1}\|_2 \|p - p^*\|_2.$$

Proof. Both probability vectors sum to one, so the constant component cancels. Cauchy–Schwarz gives the bound. Choosing $a = K^{-1} \sum_i v_i$ minimizes its coefficient.

Proposition D.2 (decision regret). Write $c_d = (C(d, a_i))_{i=1}^K$, assume Bayes minimizers exist, and define $d_p \in \arg \min_d c_d^\top p$ and $d^* \in \arg \min_d c_d^\top p^*$. If $D_C = \sup_{d,d'} \|c_d - c_{d'}\|_2 < \infty$, then

$$0 \leq c_{d_p}^\top p^* - c_{d^*}^\top p^* \leq D_C \|p - p^*\|_2.$$

Proof. Add and subtract the predicted risk difference. Optimality under p makes $(c_{d_p} - c_{d^*})^\top p \leq 0$. The remaining term is bounded by Cauchy–Schwarz. Under Proposition A.1 or the compatibility condition in A.2, substitution bounds regret by $D_C \sqrt{(R(p) - R(p^*)) / \lambda_B}$.

E Attention and head complexity

Let observation length be s , question lengths q_f , and candidate lengths $c_{f,i}$. Counting the visible key positions for every real query token gives

$$E_{\text{tree}} = s^2 + \sum_f q_f(s + q_f) + \sum_{f,i} c_{f,i}(s + q_f + c_{f,i}).$$

An attention implementation that visits these blocks has score-and-value arithmetic proportional to $E_{\text{tree}}d$ per layer, where d is the total attention width. A dense masked implementation computes over L^2 token pairs. This distinction determines how the tree’s logical sparsity translates into execution cost.

For hidden width d and utility rank r , pooling over disjoint spans costs $O(Ld)$, shared linear projections cost $O((F + N)dr)$, and gathered utility products cost $O(Nr)$. A dense span-indicator contraction additionally incurs work over its stored zero entries. The arithmetic cost using disjoint-span pooling is

$$T_{\text{head}} = O(Ld + (F + N)dr + Nr).$$

F Vector AIPW and its remainder

Assume consistency, conditional exchangeability, and overlap. Then $\mathbf{m}_t(x) = \mathbb{E}[\mathbf{Y} \mid T = t, X = x]$ identifies the treatment-specific outcome probabilities. Condition on nuisance estimates learned on an independent training fold. Write $\tilde{e}$ and $\tilde{\mathbf{m}}_t$ for those estimates, and $\boldsymbol{\psi}$ for the AIPW summand in Section 2.6.

Proposition F.1 (conditional remainder). Let $\boldsymbol{\delta}_t = \tilde{\mathbf{m}}_t - \mathbf{m}_t$. Direct conditional expectation gives

$$\mathbb{E}[\boldsymbol{\psi} \mid X] - (\mathbf{m}_1 - \mathbf{m}_0) = (\tilde{e} - e) \left(\frac{\boldsymbol{\delta}_1}{\tilde{e}} + \frac{\boldsymbol{\delta}_0}{1 - \tilde{e}} \right).$$

Proof. Substitute $\mathbb{E}[T\mathbf{Y} \mid X] = e\mathbf{m}_1$ and $\mathbb{E}[(1 - T)\mathbf{Y} \mid X] = (1 - e)\mathbf{m}_0$ into the score. Collect the coefficients of each nuisance error. The first-order terms cancel, leaving the displayed product.

If $\eta \leq \tilde{e} \leq 1 - \eta$ for $\eta > 0$, each outcome coordinate has integrated remainder bounded by

$$|\text{Rem}_k| \leq \eta^{-1} \|\tilde{e} - e\|_{L^2} (\|\boldsymbol{\delta}_{1,k}\|_{L^2} + \|\boldsymbol{\delta}_{0,k}\|_{L^2}).$$

The remainder vanishes when either nuisance component is exact, yielding double robustness. With cross-fitting, consistent nuisance estimates, fixed outcome dimension, bounded overlap, and the displayed product of errors of order $o_p(n^{-1/2})$, the score admits the usual asymptotically linear treatment-effect estimator [16; 26].

When both fitted outcome vectors lie on the simplex, every score summand also satisfies $\mathbf{1}^\top \boldsymbol{\psi} = 0$: the treatment-distribution difference and both residual vectors have zero coordinate sum. Thus the estimated distributional effect preserves total probability mass exactly.

G Entropy and information gain

Proposition G.1 (entropy bounds). For a distribution p on $K \geq 2$ outcomes, with $0 \log 0 = 0$,

$$0 \leq H(p) \leq \log K.$$

The lower bound is attained by point masses; the upper bound is attained uniquely by the uniform distribution.

Proof. Every term $-p_i \log p_i$ is nonnegative, with equality for all terms precisely at a point mass. For the uniform distribution u_K ,

$$D_{\text{KL}}(p \| u_K) = \log K - H(p) \geq 0.$$

Equality in KL holds precisely when $p = u_K$. Normalized entropy therefore lies in $[0, 1]$. If the outcome variable Z refines Y , the chain rule gives $H(Z) = H(Y) + H(Z | Y)$. The semantic support is thus part of the uncertainty definition.

Proposition G.2 (expected information gain). Fix context c and a finite-observation experiment e . Let $r_e(y, o | c)$ be a joint distribution with target marginal $p_\theta(y | c)$. Form the posterior from that same joint distribution. Then

$$\begin{aligned} \text{IG}(e | c) &= H_\theta(Y | c) - \mathbb{E}_{O \sim r_e} H_{r_e}(Y | O, c) \\ &= \sum_{y, o} r_e(y, o | c) \log \frac{r_e(y, o | c)}{p_\theta(y | c) r_e(o | c)} \\ &= I_{r_e}(Y; O | c) \\ &= \mathbb{E}_{O \sim r_e} D_{\text{KL}}(r_e(\cdot | O, c) \| p_\theta(\cdot | c)). \end{aligned}$$

Proof. Expand the two entropies and use $\sum_o r_e(y, o | c) = p_\theta(y | c)$ to combine their terms. Substituting $r_e(y, o | c) = r_e(o | c) r_e(y | o, c)$ gives the expected KL expression. Nonnegativity of KL and conditional entropy yields

$$0 \leq \text{IG}(e | c) \leq H_\theta(Y | c).$$

Zero information gain means the observation and target are independent under the specified joint law. Maximum information gain occurs when the observation determines the target almost surely. An individual observation can increase posterior entropy; averaging over the experiment’s outcomes gives the nonnegative information gain above.

The likelihood construction in Section 2.2 ensures the common target marginal by normalization. It treats experiments as acquiring evidence about a fixed target. Experiments that change the target distribution require a treatment-indexed outcome model, as developed in Section 2.6. For continuous observations, the same expected-KL identity holds with sums over observations replaced by integrals under the corresponding probability law.

H Drone-control experimental protocol

The probe is the first drone-control test scene. Its candidate set and all state fields except policy variable j remain fixed. The response reported in Section 4.5 is the larger total-variation distance over the two settings V_j :

$$D_j = \max_{v \in V_j} \frac{1}{2} \sum_{k=1}^K |p_\theta(y_k | x_{j \leftarrow v}) - p_\theta(y_k | x)|.$$

Here $x_{j \leftarrow v}$ denotes the modified scene. The paired settings are: imminent-collision threshold 0.1, 30 s; braking deceleration 0.5, 20 m/s²; caution threshold 0.2, 60 s; safety margin 0.01, 5 m; failed-scan count 0, 99; and reaction time 0.01, 3 s.

The 1,049-token scene fits in the 16,384-token state window, including its control-policy block and obstacle array; all six probes alter visible fields. Sequential fusion starts from uniform probabilities and combines predictions at imminent-threshold multipliers 0.5, 1, 8. The entropy-weighted update in Section 2.6 and an equal-weight update use the same predictions.